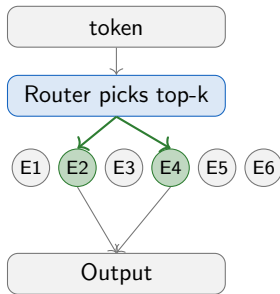


Dynamically Identifying a Safe/Harmful Subset of Experts in Mixture-of-Experts (MoE) Models

MoE LLMs: many small experts, pick a few per word

- Each transformer layer has **many small FFN experts** (often 64 or 512).
- A **router** looks at the current word and selects only a **few experts** (e.g. top-8) to contribute.
- The final output is the weighted sum of the selected experts.



router picks 2 of 6 experts here

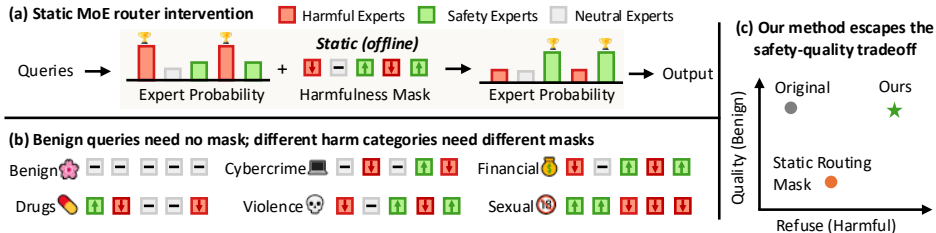
Prior work: SteerMoE (ICLR 2026) — identifies a safe-team

Observation (on **harmful inputs**). When the model **refuses**, the router picks certain experts more often; when it **complies**, a different set is picked.

SteerMoE counts, for each expert, how often it is picked during refusals vs. compliances:

- picked much more during refusal $\Rightarrow$ **safe expert** (goes into A^+);
- picked much more during compliance $\Rightarrow$ **unsafe expert** (goes into A^-).

At runtime: on every token, A^+ is forced into the router's top- k — including on benign queries, which hurts their quality.



We want to (1) skip intervention on benign queries, (2) boost only the experts actually about to be used on harmful ones.

Technique 1: a query-level “is this harmful?” gate $\lambda(q)$

Insight. Intervene *only when*, and *as much as*, the query looks harmful and the extent.

How. Build a linear “harmful-vs-benign” direction $\hat{d}_{\text{refuse}}$ (mean-diff of mid-layer hidden states on harmful vs benign prompts, offline), then project each query onto it:

$$\lambda(q) = \text{ReLU}(h(q) \cdot \hat{d}_{\text{refuse}})$$

Why it works.

- Benign $\Rightarrow \lambda=0 \Rightarrow$ defense silent $\Rightarrow$ no benign-quality loss.
- Risk grows $\Rightarrow \lambda$ grows $\Rightarrow$ defense scales naturally — no hand-tuned threshold.

<i>OLMoE-1B-7B-Instruct</i>		
	Harmful refused	Benign refused
No defense	70%	0%
SteerMoE	85%	13.3%
+ gate	85%	0%

$n = 20$ harmful + 15 benign prompts, judged by Qwen2.5-14B
(safe if score ≥ 50).

Qwen3-Next-80B (under a jailbreak attack)

	Harmful refused	Benign score
Clean (no attack)	80%	89.3
Under attack	0%	89.3
+ gate	100%	91.0

$n = 20$ harmful prompts under jailbreak; benign score on MT-Bench subset.

Technique 2: per-expert harmfulness direction v_e

Problem. SteerMoE uses a *fixed, binary* safe-team A^+ . Too coarse — ignores which experts are actually useful for *this* query.

Idea. For each expert e , learn a **harmfulness direction** v_e by asking the model: “*which input direction would both raise e ’s routing score AND make the final answer a refusal?*”

$$v_e = \nabla_h [\text{gate_logit}_e(h) + \beta \cdot \log P(\text{refuse} \mid h)]$$

Per-query, per-expert bias added to the router gate logit:

$$\text{bias}_e(q, t) = \alpha \cdot \lambda(q) \cdot \langle h(q, t), v_e \rangle$$

- $\text{sign} > 0$: expert helps refusal here $\Rightarrow$ **promote**;
- $\text{sign} < 0$: expert hurts refusal here $\Rightarrow$ **suppress**;
- ≈ 0 : irrelevant expert $\Rightarrow$ left alone.

One formula, automatically promotes safe experts *and* suppresses unsafe ones — per query, continuously.